

# Bharath Raj Nagoor Kani

✉ [bharathrajn98@gmail.com](mailto:bharathrajn98@gmail.com) •  [GitHub](#) •  [Google Scholar](#) •  [bharathrajn.com](#)

## Education

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### Cornell University

*Ph.D. in Computer Science*

Advisor: Prof. Noah Snavely

Aug 2024 – Present

### Carnegie Mellon University

*M.S. in Robotics*

Advisor: Prof. Shubham Tulsiani

Aug 2022 – May 2024

### Sri Sivasubramaniya Nadar College of Engineering

*B.E. in Electronics and Communication Engineering*

Affiliated to Anna University

June 2015 – Apr 2019

## Experience

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### Google DeepMind

*Student Researcher*

May 2025 – Aug 2025

Explored using 3D reconstruction transformers and video generative models for motion blur and defocus blur removal.

### Intel Corporation

*Research Scientist Intern*

May 2024 – Aug 2024

Explored memory efficient learning based 3D representations for real-time rendering.

### Siemens Digital Industries Software

*Engineering Services Engineer*

Jan 2022 – July 2022

*Associate Engineering Services Engineer*

May 2019 – Jan 2022

Built models, algorithms and systems for myriad autonomous driving projects. (see projects section for more details)

## Publications

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### [4] G3T Up! Gravity Aligned Coordinate Frames Simplify Pointmap Processing.

Bharath Raj Nagoor Kani, Noah Snavely

arXiv 2026

[\[Project\]](#) / [\[Paper\]](#) / [\[Code\]](#)

Shows that anchoring predictions to gravity removes a key ambiguity in pointmap-based 3D reconstruction, improving accuracy across several downstream tasks.

### [3] Flat-Pack Bench: Evaluating Spatio-Temporal Understanding in Large Vision-Language Models through Furniture Assembly.

Aditya Chetan, Eric Cai, Peeyush Kushwaha, Bharath Raj Nagoor Kani, Utkarsh Mall, Qianqian Wang, Noah Snavely, Bharath Hariharan

CVPR 2026

[\[Project\]](#) / [\[Paper\]](#) / [\[Code\]](#)

Proposes a benchmark that uses multi-step furniture assembly to test whether vision-language models can reason jointly across space and time.

### [2] UpFusion: Novel View Diffusion from Unposed Sparse View Observations.

Bharath Raj Nagoor Kani, Hsin-Ying Lee, Sergey Tulyakov, Shubham Tulsiani

ECCV 2024

[\[Project\]](#) / [\[Paper\]](#) / [\[Code\]](#)

Synthesizes novel views and infers 3D representations of an object from a sparse set of reference images *without* corresponding pose information.

### [1] Single Image Haze Removal Using a Generative Adversarial Network.

Bharath Raj N., Venkateswaran N.

WiSPNET 2020

[\[Paper\]](#) / [\[Code\]](#)

Proposes a conditional GAN based architecture to remove haze from images.

# Projects

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## Progressive Photon Mapping

Implemented Progressive Photon Mapping in C++ for the Physics-based Rendering course (15-668) at CMU by extending the renderer to handle light paths of type  $L(S^+)D(S^+)$  for caustic effects. [\[Report\]](#)

## Open Source Contributions to Kornia

Contributed enhancements and fixes to Kornia, an open source differentiable computer vision library for PyTorch, including a Direct Linear Transform (DLT) based Perspective-n-Point (PnP) solver.

## Autonomous Driving Projects at Siemens

- **Ego-Lane Estimation & ROS Perception Toolchain:** Leveraged 3D geometry, machine learning and state estimation to build a fast, robust ego-lane estimation and tracking system, and implemented core components of a ROS-based perception toolchain.
- **Generative Vehicle Trajectory Prediction:** Researched generative adversarial networks with a structured latent space for predicting future ego-vehicle trajectories.
- **Maximum Entropy Inverse RL:** Implemented maximum entropy inverse reinforcement learning algorithms to model highway driving styles from expert demonstrations.

# Teaching & Service

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## Mentoring

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|--------------------------------------------------------------------------|---------------------|
| • <b>Bowers Undergraduate Research Experience</b> , Cornell University   | May 2026 – Present  |
| • <b>Undergraduate AI Mentoring Program</b> , Carnegie Mellon University | Oct 2023 – May 2024 |
| • <b>Google Code-In Mentor</b> , CloudCV                                 | Oct 2018 – Dec 2018 |

## Reviewing

- **CVPR** (2024–2026), **ECCV/ICCV** (2024–2026), **BMVC** (2026)

## Graduate Teaching Assistant

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|----------------------------------------------------------------------|-------------|
| • <b>CS 5670: Introduction to Computer Vision</b> , Cornell Tech     | Fall 2025   |
| • <b>16-825: Learning for 3D Vision</b> , Carnegie Mellon University | Spring 2024 |

# Honors & Awards

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|------------------------------------|------|
| <b>Outstanding Reviewer</b> , ECCV | 2026 |
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## Hackathon Awards

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|--------------------------------------------------------------|----------|
| • <b>People’s Choice Award</b> , Yet Another Hackathon, SVCE | Aug 2018 |
| • <b>Runner Up</b> , AWS Deep Learning Hackathon, IIT Madras | Jan 2018 |